

EXPANSION too few masks → grow canvas

t = T · initial canvas

```
int <|mask|> = 0;
```

only 1 mask reserved — target needs 2 BPE tokens

predict

step k · model prediction

```
int <|expand|> = 0;
```

the mask is predicted as <|expand|>

rule

step k · after substitution rule

```
int <|mask|><|mask|> = 0;
```

rule: <|expand|> → two <|mask|> (canvas + 1)

denoise

t = 0 · converged

```
int activeCount = 0;
```

the two masks denoise to regular BPE tokens

DELETION too many masks → shrink canvas

t = T · initial canvas

```
int <|mask|><|mask|><|mask|> = 0;
```

3 masks reserved — target needs only 1 BPE token

predict

step k · model prediction

```
int n<|delete|><|delete|> = 0;
```

one mask → "n"; two masks → <|delete|>

rule

step k · after substitution rule

```
int n = 0;
```

rule: <|delete|> → removed (canvas - 2)

denoise

t = 0 · converged

```
int n = 0;
```

canvas length now matches the target

■ <|mask|>

■ <|expand|> → two <|mask|>

■ <|delete|> → removed

■ denoised token